

TUCUMCARI, NM, USA

WMO: 723676

Lat: 35.182N

Lon: 103.603W

Elev: 1239

StdP: 87.30

Time Zone: -7.00 (NAM)

Period: 00-19

WBAN: 23048

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.1	(c) -8.4	(d) -19.1	(e) 0.8	(f) 5.8	(g) -17.2	(h) 1.0	(i) 5.1	(j) 15.6	(k) 14.0	(l) 14.2	(m) 13.2	(n) 2.8	(o) 60	(p) 0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	38.3	17.4	37.1	17.9	35.7	17.9	20.8	30.7	20.4	30.3	19.9	29.9	6.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	15.7	22.7	17.9	15.0	22.6	17.3	14.4	22.5	66.2	30.5	63.9	30.2	62.2	29.4	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 14.0	(b) 11.9	(c) 10.7	(e) -16.8	(f) 40.7	(g) 3.9	(h) 1.2	(i) -19.6	(j) 41.6	(k) -21.8	(l) 42.3	(m) -24.0	(n) 42.9	(o) -26.8	(p) 43.8		
(a) 14.0	(b) 11.9	(c) 10.7	(e) -17.3	(f) 22.3	(g) 3.6	(h) 1.9	(i) -19.8	(j) 23.7	(k) -21.9	(l) 24.8	(m) -24.0	(n) 25.8	(o) -26.6	(p) 27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.4	3.8	5.7	10.4	14.9	19.6	25.6	27.2	26.1	22.2	15.4	8.9	3.9	(6)
(7)		DBStd	9.58	5.28	5.70	5.32	4.72	4.56	3.30	2.55	2.76	3.54	4.82	5.34	5.50	(7)
(8)		HDD10.0	706	199	139	59	13	2	0	0	0	0	12	82	199	(8)
(9)		HDD18.3	2069	449	354	248	119	40	1	0	0	10	114	285	448	(9)
(10)		CDD10.0	2661	8	18	72	160	299	469	534	499	366	178	48	9	(10)
(11)		CDD18.3	982	0	0	2	17	78	219	276	241	126	23	1	0	(11)
(12)		CDH23.3	12506	1	16	133	422	1192	2832	3417	2696	1386	367	43	1	(12)
(13)		CDH26.7	6618	0	2	28	132	564	1637	1988	1489	648	126	4	0	(13)
(14)	Wind	WSAvg	4.8	4.5	4.9	5.2	5.8	5.2	5.1	4.4	4.1	4.4	4.5	4.7	4.6	(14)
(15)	Precipitation	PrecAvg	377	6	10	10	23	37	43	83	68	43	25	15	7	(15)
(16)		PrecMax	694	30	33	51	87	128	108	175	177	127	113	73	27	(16)
(17)		PrecMin	223	0	0	0	0	4	2	11	10	4	0	0	0	(17)
(18)		PrecStd	109	7	10	11	21	33	31	43	54	32	27	19	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.3	25.7	29.7	32.4	36.3	40.6	40.3	38.8	36.3	32.6	27.0	22.3	(19)
MCWB			8.4	10.2	11.3	12.5	15.3	16.4	17.8	18.1	17.3	14.8	11.0	8.8	(20)	
2%		DB	19.3	22.4	27.0	30.0	34.2	38.6	38.1	37.2	34.6	30.0	24.5	19.0	(21)	
		MCWB	7.4	8.5	10.4	11.7	14.4	16.0	18.6	18.7	17.0	14.3	10.4	7.6	(22)	
5%		DB	16.7	19.5	24.3	27.9	32.4	37.1	36.8	35.9	32.9	27.7	21.7	16.4	(23)	
		MCWB	6.3	7.3	9.6	11.4	14.0	16.4	18.8	18.8	17.0	13.6	9.6	6.4	(24)	
10%		DB	13.7	16.6	21.4	25.9	30.2	35.0	35.4	34.0	31.0	24.8	18.7	13.6	(25)	
		MCWB	5.0	6.1	8.6	11.0	13.8	16.9	18.7	18.8	16.8	13.0	8.5	5.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	10.8	13.5	15.2	19.0	21.2	21.7	21.7	20.5	19.2	13.6	10.1	(27)
MCDWB			19.1	22.3	22.2	22.7	28.3	30.4	31.3	30.8	29.4	24.7	20.3	18.5	(28)	
2%		WB	8.0	9.3	11.6	14.1	17.6	20.1	20.8	20.8	19.6	17.3	11.8	8.4	(29)	
		MCDWB	17.7	20.3	22.7	22.8	27.2	30.3	31.3	30.4	27.5	22.5	20.5	18.0	(30)	
5%		WB	6.7	8.0	10.4	13.0	16.6	19.4	20.4	20.3	18.9	15.6	10.5	6.9	(31)	
		MCDWB	15.6	18.1	22.1	23.3	26.1	29.9	30.8	30.0	27.1	22.8	19.8	15.3	(32)	
10%		WB	5.3	6.7	9.3	11.8	15.5	18.8	19.9	19.9	18.3	14.3	9.2	5.5	(33)	
		MCDWB	13.1	15.9	20.4	23.4	24.7	29.3	30.4	29.6	26.6	22.7	17.7	13.1	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.8	16.4	17.1	17.2	16.6	16.7	15.2	14.8	14.9	15.5	15.7	14.7	(35)
MCDBR			19.5	20.8	21.1	19.9	19.4	18.8	17.0	16.8	17.2	18.8	19.5	18.8	(36)	
MCWBR			10.3	10.2	9.4	8.2	7.5	5.9	4.6	4.4	5.5	7.4	9.0	10.0	(37)	
5% WB		MCDWB	18.3	19.1	19.0	16.8	16.6	16.1	15.0	14.8	14.2	14.4	17.0	17.6	(38)	
		MCWBR	10.0	9.7	8.8	8.1	7.1	5.5	4.2	4.3	4.9	6.7	8.3	9.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.250	0.257	0.282	0.310	0.323	0.361	0.380	0.363	0.331	0.292	0.266	0.253	(40)	
taud		2.606	2.594	2.521	2.437	2.457	2.374	2.378	2.431	2.470	2.566	2.571	2.595	(41)		
Ebn at Noon		980	1005	997	975	959	917	897	910	929	948	948	955	(42)		
Edh at Noon		73	83	98	113	112	122	121	112	102	85	75	70	(43)		
(44)	All-Sky Solar Radiation	RadAvg	3.25	4.17	5.43	6.71	7.47	7.87	7.46	6.75	5.77	4.47	3.46	2.82	(44)	
(45)		RadStd	0.24	0.32	0.39	0.37	0.41	0.33	0.31	0.30	0.29	0.40	0.26	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.61	N/A	N/A	+0.89	N/A	N/A	N/A	N/A	+197	+134
(47) Regional (11 neighbors)		+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+76	+49

Nomenclature: See separate page